

Qatar-2b

R-0959

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_170408 · created 2026-07-29T15:44:21+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457851.90278 +/- 0.00043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.065 +/- 0.035
Transit depth (Rp/Rs)^2	0.43 +/- 0.46 [%]
Orbital Inclination (inc)	84.07 +/- 1.73
Airmass coefficient 1 (a1)	1.043 +/- 0.036
Airmass coefficient 2 (a2)	-0.026 +/- 0.013
Scatter in the residuals of the lightcurve fit is	3.44 %
Best Comparison Star	None
Optimal Aperture	8.86
Optimal Annulus	16.98
Transit Duration (day)	0.052 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.065 $\pm$ 0.035 vs 0.16327 (deviation 2.81 σ)
Duration fit vs published	0.052 d vs 0.0767 d (deviation 1.9 σ)
Mid-time O-C	-0.7 min
Depth SNR	1.9
Residual scatter	3.44 %

Light curve

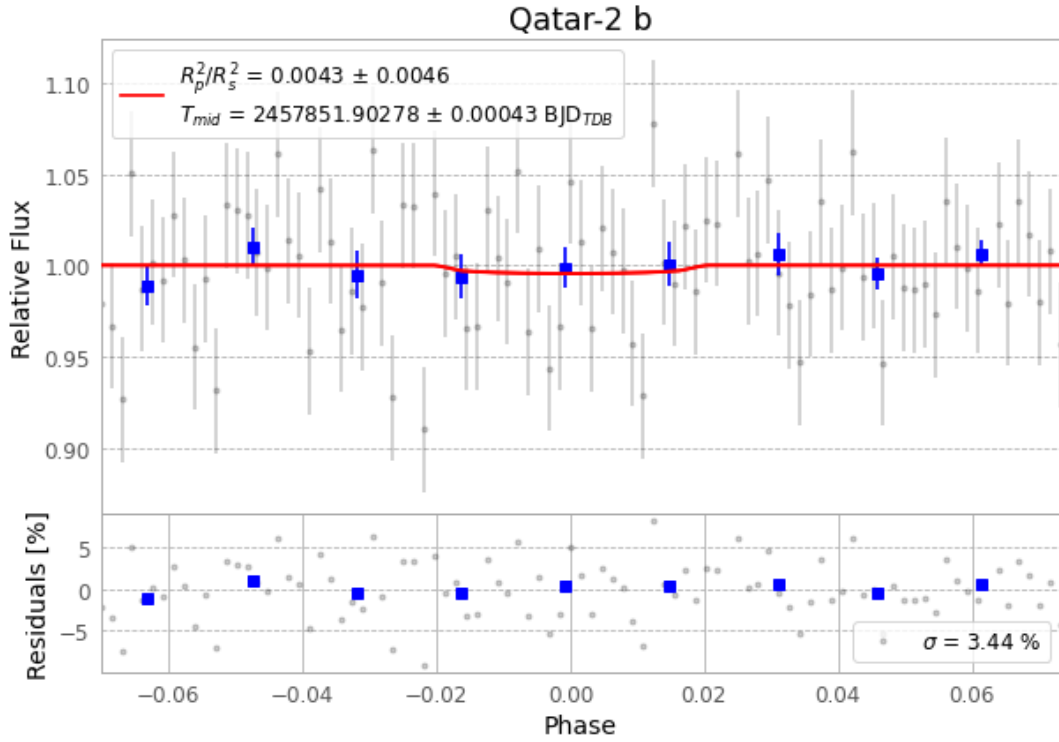


Figure 1 — FinalLightCurve_Qatar-2 b_2017-04-08.png (SHA-256 7cfd2575aad3c394...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [411, 447], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a550c4b79192fcfd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

93 FITS frames (61 MB), Qatar-2170408071512.FITS → Qatar-2170408115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-170408.log (fdbcd0c7a2ad...), FinalLightCurve_Qatar-2 b_2017-04-08.png (7cfd2575aad3...), FinalLightCurve_Qatar-2 b_2017-04-08.csv (7607590ac587...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 15:44Z.